

Deepak Mallampati

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Kent, OH -- F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) shipping production agentic LLM systems with grounding controls and audit-ready reasoning. Track record in HIPAA-conscious healthcare workflows: RAG (~**35%** retrieval precision), agentic LLM workflows with structured reasoning (>**30%** hallucination reduction), and predictive ML (**15-20%** recall gain on high-risk patient categories). Built **Agentic Healthcare Claims** pipeline with schema-validated JSON contracts between agents to eliminate cascading hallucinations -- directly applicable to multi-turn payer-call structured capture. Independently shipped **Manga Lens** Chrome extension (4 AI providers, 29 sites). Comfortable with REST APIs, observability, and ambiguous early-stage scope.

CORE COMPETENCIES

Agentic LLM Systems . Production RAG . Structured Outputs & Guardrails . Evaluation & QA Infrastructure . Python / FastAPI / REST APIs . Healthcare AI & HIPAA . EHR Integrations & Prior Auth . Observability & Monitoring . CI/CD & Production Discipline . End-to-End Ownership . Ambiguity / Early-Stage Ship

EXPERIENCE

Progress Solutions Inc.

Jul 2025 -- Present

Jr. AI/ML Engineer Trainee -- USA | Healthcare Technology, AI Consulting

- Developed **agentic LLM workflows** for multi-step healthcare queries (eligibility checks, care workflow navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; improved agent stability ~**25%** and reduced hallucination rate >**30%**.
- Built **RAG systems** for clinical knowledge retrieval; recursive semantic chunking + transformer embeddings improved precision ~**35%** and reduced irrelevant retrieval >**30%**; response alignment exceeded **90%**.
- Packaged LLM/ML inference as **FastAPI/Flask** RESTful services containerized with Docker; structured logging + load simulation cut post-deployment defects ~**30%**.
- Built evaluation + observability tooling: lineage docs, audit trails, system-limitation docs for non-technical clinical stakeholders -- enabled adoption in HIPAA-conscious environment.
- Shipped predictive ML pipelines (scikit-learn, XGBoost) for patient no-show and care engagement; improved recall **15-20%** on high-risk categories.

Suvidha Foundation

Jan 2022 -- Mar 2022

Machine Learning Engineer -- Hyderabad, India | Nonprofit Tech, Applied NLP

- Built transformer-based video summarization for 5,000+ sessions; reduced manual review time **60-70%**; deployed as Flask API for non-technical users.
- Implemented document Q&A with RAG; reduced hallucinated outputs ~**30%**.

Energy Solutions International (Emerson)

Jun 2022 -- Apr 2023

Database & DevOps Performance Engineer (Intern) -- Hyderabad, India | Enterprise ERP

- Built CI/CD pipelines with Jenkins; reduced deployment errors >30%; improved release cycle ~35-40%.
- Designed Grafana observability dashboards for production SQL systems; cut incident recurrence ~25%.

PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence

Multi-agent pipeline (Intake -> Validation -> Consistency Analysis -> Duplicate Detection -> Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations -- direct analog to multi-turn payer-call structured capture. RAG-grounded CPT/ICD validation against vector-indexed policy docs; ANN similarity for duplicate detection; explainable risk scoring with audit-ready reasoning traces.

Python, LangChain, vector search, FastAPI, Docker, healthcare ontology

Manga Lens shipped -- Chrome Web Store

Solo-built Chrome extension integrating **4 AI vision providers** via provider-agnostic abstraction (Claude, GPT-4o, Ollama). Multi-section capture pipeline, 7-day cache, 29 site selector configs. Full ship discipline: privacy policy, scoped permissions, Chrome review.

TypeScript, Manifest V3, content scripts, multi-provider abstraction

Dream Decoder -- AI Multimodal Platform

Full-stack FastAPI + React/TS app with coordinated multimodal APIs. Intermediate structured prompt transformation layers improved contextual alignment **~30%** and first-pass success **~25-30%**.

FastAPI, React, TypeScript, Perplexity Sonar, Replicate

EDUCATION

Master's Degree -- Kent State University, Ohio, USA

Aug 2023 -- May 2025

TECHNICAL SKILLS

Languages & Frameworks: Python, TypeScript, JavaScript, React, FastAPI, Flask, SQL (PostgreSQL, T-SQL, SQLite)

AI/LLM & Agentic: LLM APIs (Claude, GPT-4o, Gemini, Ollama), LangChain, LlamaIndex, RAG, multi-agent workflows, prompt engineering, structured outputs, evaluation pipelines, guardrails, grounding

Healthcare: EHR data, HIPAA-conscious governance, de-identification, claims processing (CPT/ICD), audit trails

Developer Tooling: Cursor, Claude Code, Docker, CI/CD (Jenkins), Grafana, REST APIs, MLOps/LLMOps, observability/logging

Work Authorization: F-1 OPT -- authorized to work in the United States now; future sponsorship required